

De Novo Binder Docking Album

Top 5 designed miniprotein binders — A β 42 (Alzheimer's) target

BindCraft campaign • ranked by Stage-5 composite score • target 9CO4 (receptor-bound A β 42 fibril)

Each design is an AlphaFold2-backpropagation binder docked onto the A β 42 fibril target. Left panel: the helical binder (blue) nestled against the target molecular surface (wheat). Right panel: the binding interface — contacting side chains as sticks, with the designed hotspot cluster (A β 42 residues 10–16) in orange. All 5 engage all 6 hotspots.

#	Design (scaffold_mpnn)	Len	i-pTM	i-pAE	dG	SC	Score	Iface res (bind/tgt)
1	l82_s967366_mpnn11	82	0.79	0.17	-71.8	0.76	0.793	16/17
2	l89_s578974_mpnn11	89	0.80	0.19	-70.8	0.68	0.733	18/23
3	l90_s311742_mpnn16	90	0.86	0.15	-88.8	0.76	0.713	23/22
4	l71_s843399_mpnn18	71	0.80	0.18	-76.6	0.71	0.708	20/16
5	l90_s311742_mpnn3	90	0.86	0.15	-88.0	0.76	0.696	21/21

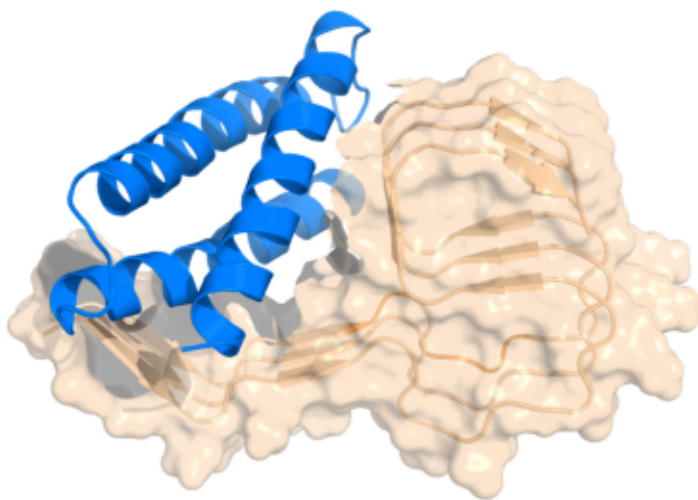
Chain A = A β 42 target (9CO4, C/E/G fibril chains) • Chain B = designed binder

Prepared 2026-07-05 • renders: PyMOL 3.1.0 (ray-traced) • ab42-binder-design

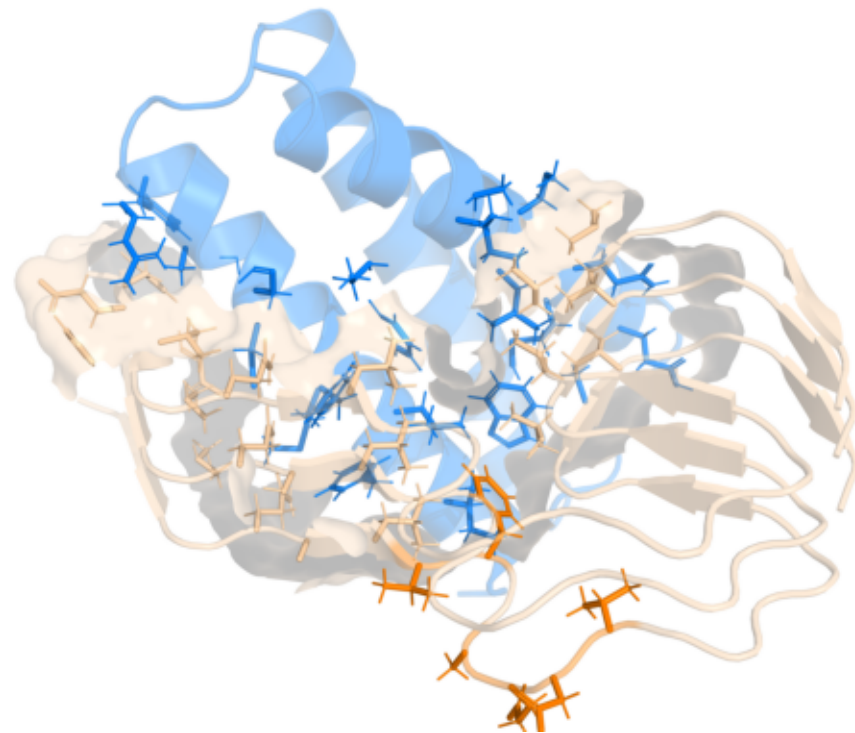
#1 l82_s967366_mpnn11

scaffold s967366 • 82 aa • i-pTM 0.79 • i-pAE 0.17 • ΔG -71.8 kcal/mol • shape-comp 0.76 • score 0.793

Overview — binder on target surface



Interface — contact side chains + hotspots

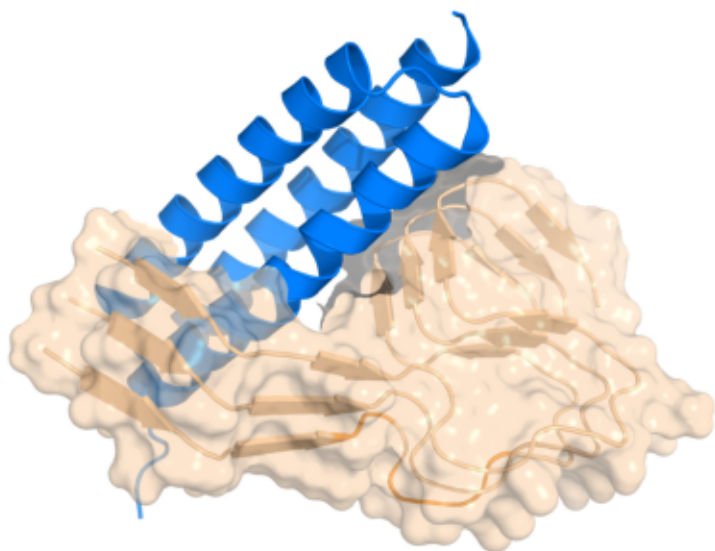


The binder buries 16 residues against 17 A β 42 residues at the interface, engaging all 6 designed hotspots (A β 42 10,11,13-16; orange). Blue = binder (chain B), wheat = A β 42 target (chain A). Predicted binding confidence i-pTM 0.79 with interface PAE 0.17 Å and Rosetta ΔG -71.8 kcal/mol — a tight, well-packed lateral docking.

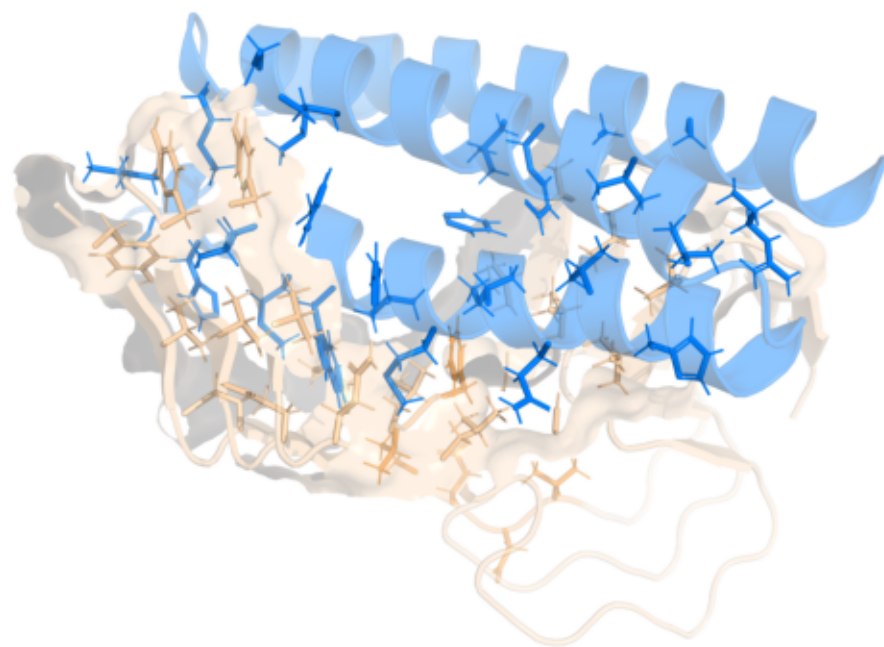
#2 l89_s578974_mpnn11

scaffold s578974 • 89 aa • i-pTM 0.80 • i-pAE 0.19 • ΔG -70.8 kcal/mol • shape-comp 0.68 • score 0.733

Overview — binder on target surface



Interface — contact side chains + hotspots

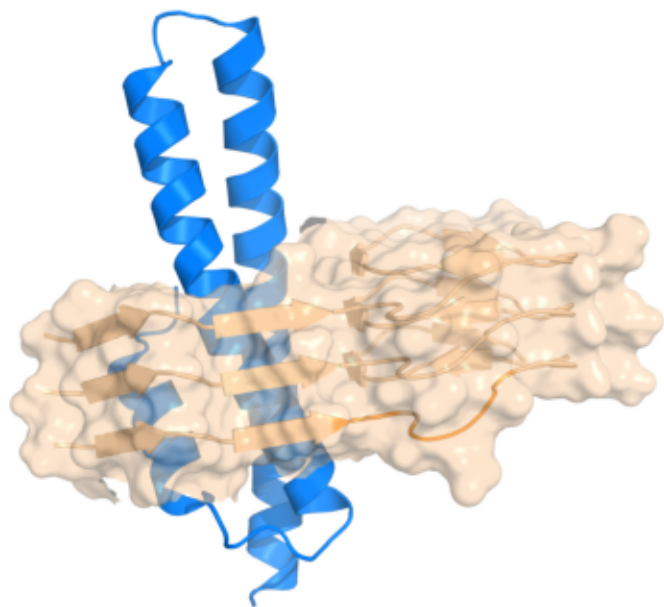


The binder buries 18 residues against 23 Aβ42 residues at the interface, engaging all 6 designed hotspots (Aβ42 10,11,13-16; orange). Blue = binder (chain B), wheat = Aβ42 target (chain A). Predicted binding confidence i-pTM 0.80 with interface PAE 0.19 Å and Rosetta ΔG -70.8 kcal/mol — a tight, well-packed lateral docking.

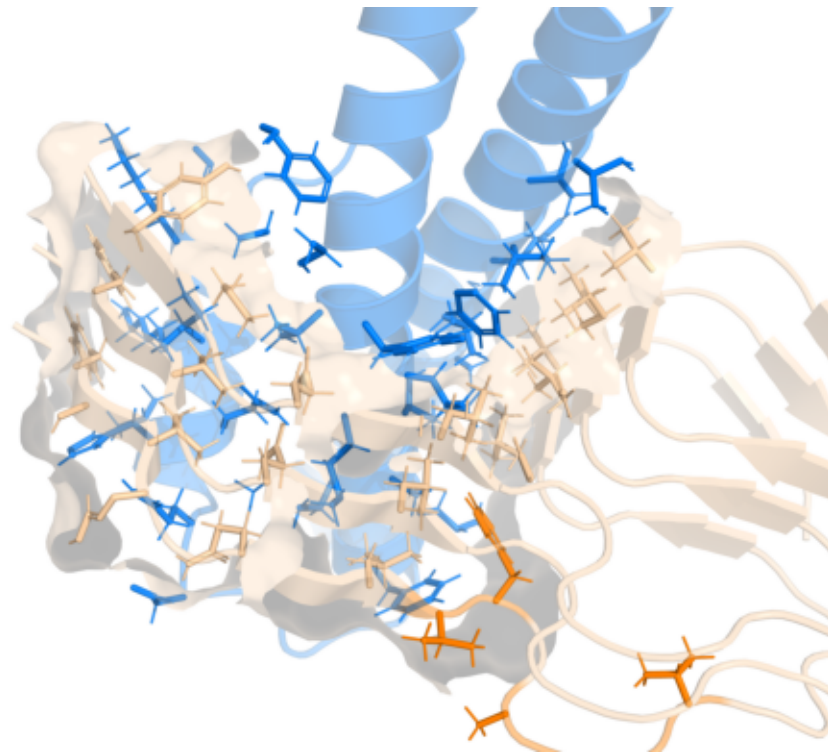
#3 I90_s311742_mpnn16

scaffold s311742 • 90 aa • i-pTM 0.86 • i-pAE 0.15 • ΔG -88.8 kcal/mol • shape-comp 0.76 • score 0.713

Overview — binder on target surface



Interface — contact side chains + hotspots

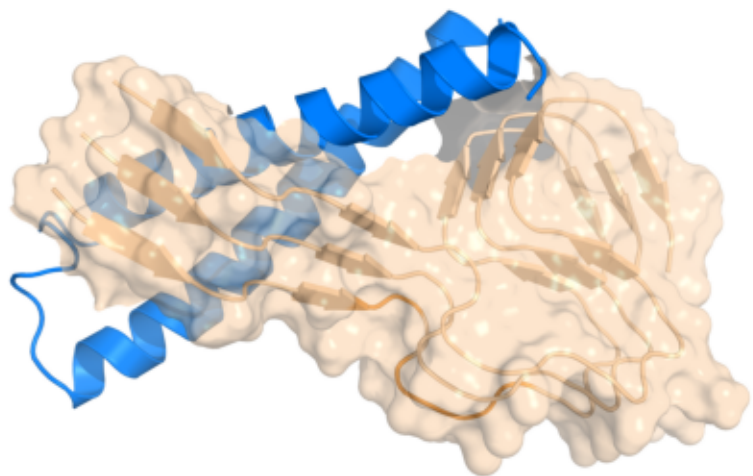


The binder buries 23 residues against 22 Aβ42 residues at the interface, engaging all 6 designed hotspots (Aβ42 10,11,13-16; orange). Blue = binder (chain B), wheat = Aβ42 target (chain A). Predicted binding confidence i-pTM 0.86 with interface PAE 0.15 Å and Rosetta ΔG -88.8 kcal/mol — a tight, well-packed lateral docking.

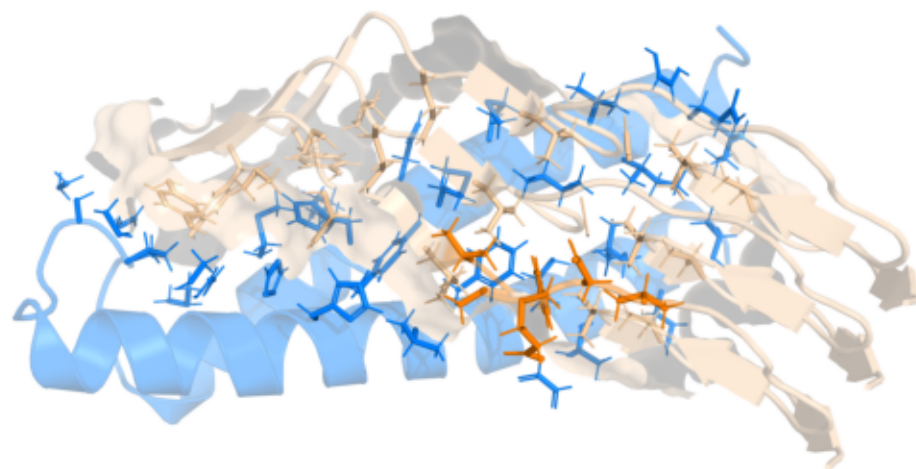
#4 l71_s843399_mpnn18

scaffold s843399 • 71 aa • i-pTM 0.80 • i-pAE 0.18 • ΔG -76.6 kcal/mol • shape-comp 0.71 • score 0.708

Overview — binder on target surface



Interface — contact side chains + hotspots

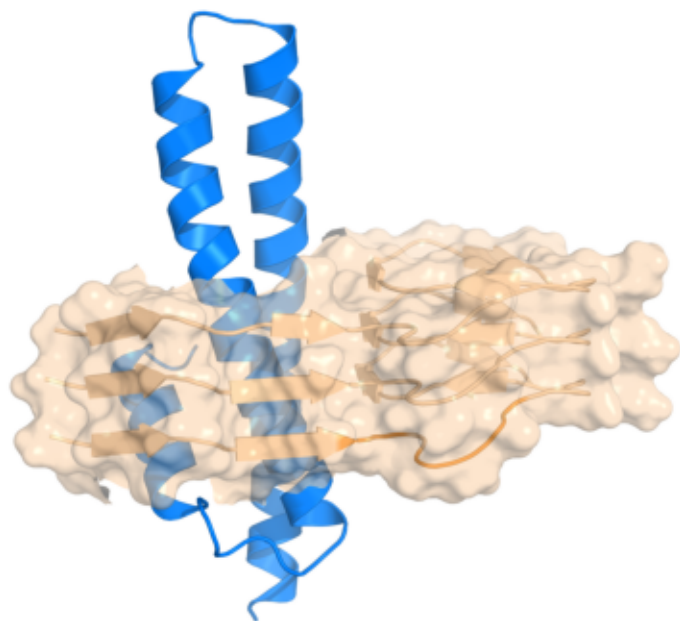


The binder buries 20 residues against 16 A β 42 residues at the interface, engaging all 6 designed hotspots (A β 42 10,11,13-16; orange). Blue = binder (chain B), wheat = A β 42 target (chain A). Predicted binding confidence i-pTM 0.80 with interface PAE 0.18 Å and Rosetta ΔG -76.6 kcal/mol — a tight, well-packed lateral docking.

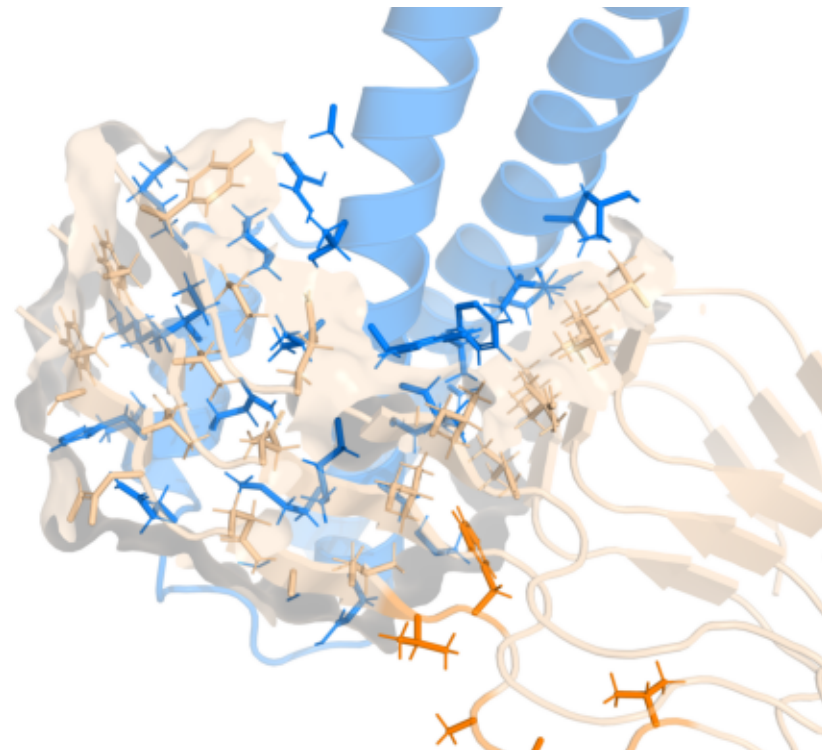
#5 I90_s311742_mpnn3

scaffold s311742 • 90 aa • i-pTM 0.86 • i-pAE 0.15 • ΔG -88.0 kcal/mol • shape-comp 0.76 • score 0.696

Overview — binder on target surface



Interface — contact side chains + hotspots



The binder buries 21 residues against 21 Aβ42 residues at the interface, engaging all 6 designed hotspots (Aβ42 10,11,13-16; orange). Blue = binder (chain B), wheat = Aβ42 target (chain A). Predicted binding confidence i-pTM 0.86 with interface PAE 0.15 Å and Rosetta ΔG -88.0 kcal/mol — a tight, well-packed lateral docking.